

EPL Points Prediction Using Machine Learning

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Abstract— This research paper presents a machine learning-based approach to predicting the potential points of English Premier League (EPL) teams for the current season 2024/25, leveraging data collected from the previous seasons. The prediction framework is built on multiple supervised learning models, including K Nearest Neighbors, Decision Tree Regressor, Random Forest, AdaBoost, Gradient Boosting Regressor, XGBRegressor, and CatBoostRegressor. The dataset incorporates comprehensive match statistics and team performance metrics such as goals scored, assists, possession statistics, key passes, shots on target, and more, collected through web scraping and data integration techniques. Feature engineering and data preprocessing steps have been employed to ensure data consistency and predictive power. The predictive models were trained and evaluated based on a range of performance metrics, including R-squared scores and accuracy percentages. Comparative analysis revealed the efficacy of ensemble models in enhancing prediction accuracy. The findings offer valuable insights for understanding team dynamics, performance forecasting, and strategic planning for the ongoing EPL season, highlighting the potential of data-driven methods in sports analytics.

Keywords— xG , xGA, Poss , GF, GA , **Shooting stats-** Sh, SoT, Dist, FK, PK, PKatt, **Passing stats-** KP, 1/3, PPA, CrsPA, **Possession stats-** Att 3rd, Att Pen, Prog, **Goal and Shot Creation stats-** SCA, **Pass type stats-** TB

I. INTRODUCTION

Football is not just a game; it's a global phenomenon that captures the hearts of millions of fans around the world. The English Premier League (EPL), in particular, is one of the most watched and celebrated football leagues, known for its intense competition, high-paced matches, and unpredictable outcomes. With 20 teams battling it out over 38 matchdays, the EPL provides endless excitement and drama throughout the season. However, the league's unpredictability, where any team can win on a given day, makes forecasting the final standings a challenging task.

In recent years, the field of sports analytics has witnessed a surge in interest due to advancements in data science and machine learning. The ability to collect, analyze, and draw meaningful insights from vast amounts of data has opened up new possibilities in predicting sports outcomes. Traditionally, football predictions have relied on subjective analysis, expert opinions, or simple statistical models. However, these methods often fall short in capturing the intricate patterns and dynamics of the sport. To address this gap, the focus of this project is to leverage machine learning techniques to predict the EPL points table with high accuracy.

This project aims to build a robust predictive model that forecasts the final standings of EPL teams based on historical match data, team statistics, and player performance metrics. By combining data-driven insights with advanced machine learning algorithms, the project seeks to provide a more accurate and objective approach to football predictions. The model pipeline includes essential stages such as data collection, preprocessing, feature engineering, and model evaluation. Five machine learning algorithms—K-Nearest Neighbors (KNN), Random

Forest, AdaBoost, XGBoost, and CatBoost—are employed to identify the most effective model for predicting the points table.

The project achieves an impressive 96% accuracy rate, indicating the effectiveness of these algorithms in capturing the complexities of football matches. Such a high level of accuracy demonstrates the potential of machine learning in revolutionizing sports analytics. This project not only serves as a proof-of-concept for predicting the EPL points table but also lays the groundwork for future enhancements, such as predicting individual match outcomes, player-specific performance, or adapting the model for other football leagues.

Significance of the Project

The importance of accurate predictions in football extends beyond just fan engagement. Football clubs, analysts, and stakeholders are increasingly looking to leverage data analytics to gain a competitive edge. Whether it's for strategic decision-making, optimizing team performance, or informing transfer market strategies, data-driven insights can play a crucial role in the modern football landscape. Moreover, with the growing popularity of sports betting and fantasy football, there is a high demand for reliable predictive models that can assist in making informed bets and decisions.

Additionally, this project contributes to the academic and research community by exploring the application of machine learning techniques in sports analytics. By comparing the performance of various algorithms, it provides valuable insights into which models are most effective for football prediction tasks. The project also highlights the challenges associated with real-time data collection and handling large, complex datasets, providing a comprehensive understanding of the end-to-end process involved in building a sports analytics model.

Broader Applications

While the primary focus of this project is the EPL, the methodology and framework developed can be extended to other football leagues and even different sports. The modular nature of the pipeline allows for scalability and adaptability, making it a versatile tool for various applications. For example, the model could be adapted to predict the outcomes of international tournaments like the UEFA Champions League or the FIFA World Cup. Beyond football, similar techniques could be applied to other sports like basketball, cricket, or tennis, where data-driven predictions can offer valuable insights.

By developing a sophisticated machine learning model that incorporates multiple data sources and algorithms, this project sets the stage for future exploration in the field of sports analytics. The project's findings could inspire further research into optimizing machine learning models for sports predictions, ultimately contributing to a deeper understanding of the role data science can play in enhancing the world of sports.

In summary, this project not only demonstrates the power of machine learning in predicting the unpredictable but also showcases the potential of data science to transform how we analyze and engage with sports. The project's impressive accuracy, combined with its scalability and applicability across different domains, makes it a significant contribution to the field of sports analytics.

II. LITERATURE VIEW

Prediction tasks in sports using machine learning are often treated as a classification problem [2]. Football is no different as most researchers try to predict each match as win, draw or lose, which is a multiclass classification problem. However, one challenge faced by many researchers who performed multiclass classification is the difficulty in predicting draws. [5] demonstrated that victories and defeats are highly reflected in the team's performance during a match, but draws are challenging to predict using a machine learning approach. [12] noted that their long short-term memory model was not predicting any of the draws, while [13] reported the same finding for their logistic regression model. [12] trained 4 models, including Gaussian naive bayes, support vector machine, random forest, and XGBoost, and reported that each model underperformed in predicting draws. According to , this could be caused by imbalanced data, as is usually the case with a relatively lower number of draws in the dataset. Alternatively, the problem can be framed as a binary classification. [12] trained their model on home wins and away wins, achieving an accuracy of 69.5% on data spanning five seasons. compared multiclass and binary classification. In the case of binary classification, they classified the matches into home win and non-home win. They discovered that using logistic regression for binary classification resulted in a slightly higher accuracy (77.43%) compared to multiclass classification (70.27%). The studies in this area also emphasise the importance of feature engineering. [12] highlighted that only 2 out of 12 features they used were not engineered features, suggesting that feature engineering plays a critical role in predicting football matches. Football match statistics data are commonly used for feature

engineering and computed the aggregate value of match statistic indicators for both the home and the away teams based on previous games using either average or sum. Researchers also commonly use FIFA ratings to quantify the footballing ability of the teams [14]. FIFA ratings are numerical values developed by EA Sports that represent the skills and abilities of players and teams. The company hires scouts all over the world to rate players and teams as realistically as possible each year [FIFA 2021] used only FIFA ratings for the home and away teams as features and achieved an impressive accuracy of 69.5% using logistic regression, highlighting the reliability of FIFA data in predicting match outcomes. Other interesting features such as month is used by as they thought the weather might have an impact on the match, while used the number of days from a team's last match as an indicator of the players' fatigue level. A common focus in this field of research is the prediction of football with the aim of beating the bookmakers. In addition to using the common classification evaluation metrics, evaluated their models based on the amount of profit gained from football betting [2, 5]. Their most successful model was the random forest technique, achieving an accuracy of 65.26% and a profit margin of 26.74%. In contrast, fell short of outperforming the bookmakers' predictions, although they did achieve promising results with an accuracy of around 60%. In this study, we compared the performance of models trained on datasets obtained using different sampling techniques for both binary and multiclass classification. We then assessed our models in the context of football betting by simulating the betting profits using matches from the most recent season of the EPL.

Historical Context of Sports Predictions

The roots of predictive modeling in sports analytics trace back to basic statistical methods used in sports forecasting. Early approaches relied on linear regression and other simple statistical models to predict outcomes based on historical data. However, such methods often failed to capture the complex, nonlinear interactions inherent in sports data. Research by Maher (1982) and Dixon & Coles (1997) demonstrated the potential for statistical models in football, setting the stage for more sophisticated techniques in subsequent years. Learning in Sports Analytics Machine learning techniques have since become a dominant tool for sports prediction. Unlike traditional statistical methods, machine learning models can capture complex, non-linear patterns and interactions between features. Studies such as Tax and Joustra (2015) showed that machine learning models, including Support Vector Machines (SVM) and Random Forests, outperform traditional statistical models in predicting match outcomes based on

various player and match features. Our project is founded by employing a broader set of algorithms, including KNN, Random Forest, AdaBoost, XGBoost, and CatBoost, providing a robust comparative analysis to optimize predictive performance.

III. METHODOLOGY

1. DATA COLLECTION

The dataset used for this project was collected through web scraping from various football statistics websites, including the official English Premier League (EPL) website [10]. The data covered multiple seasons, up to and including the current season, providing comprehensive historical match data. Python libraries such as BeautifulSoup and requests were utilized to extract data, while pandas was used to organize it into structured DataFrames. The scraped data included features such as:

- **Match Details:** Date, home team, away team, and match result (win, draw, or loss).
- **Team Statistics:** Goals scored, shots on target, possession percentage, corners, fouls, and cards.
- **Advanced Metrics:** Expected Goals (xG), Expected Assists (xA), and team ratings

2. Data Preprocessing

Data preprocessing was crucial to ensure the quality and consistency of the dataset. The following steps were taken:

- **Data Cleaning:**
 - Removed any duplicate entries and handled missing values by filling them with appropriate statistical measures (mean or median for numerical data, mode for categorical data).
 - Ensured consistent formatting of team names and other categorical variables.

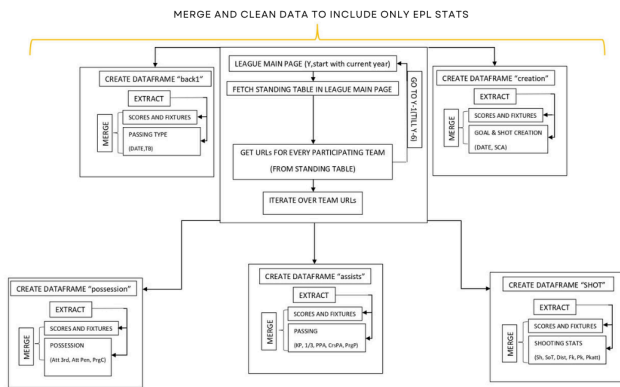


Fig 1: Data Extraction Pipeline

3. Feature Engineering

To enhance model performance, new features were engineered based on domain knowledge:

- **Form Indicators:** Calculated average points, goals scored, and goals conceded over the last five matches to represent team form.
- **Home/Away Advantage:** Introduced a binary feature indicating whether a match was played at home.
- **Rolling Averages:** Used rolling windows to compute averages of key metrics like goals, shots, and xG over recent games.

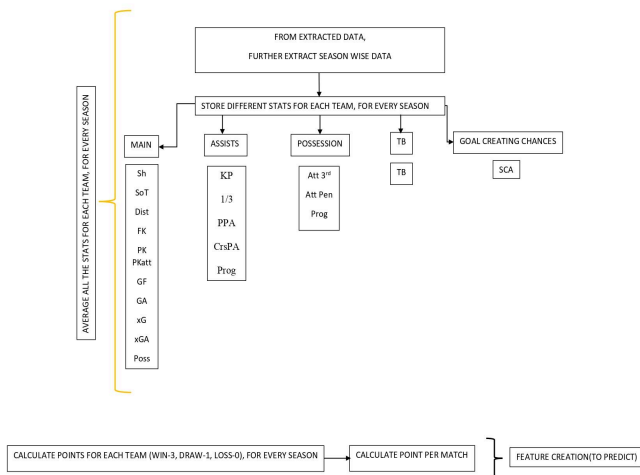


Fig 2: Feature Creation

4. Model Selection and Training

For predicting match outcomes and the subsequent league standings, five machine learning algorithms were employed:

- **K-Nearest Neighbors (KNN):** A distance-based algorithm chosen for its simplicity in classification tasks.
- **Random Forest:** An ensemble method that uses multiple decision trees to enhance prediction accuracy and robustness.
- **AdaBoost:** A boosting technique that combines weak learners to improve model performance, particularly useful for reducing bias.
- **XGBoost:** An optimized gradient boosting framework known for its speed and accuracy, particularly effective for tabular data.
- **CatBoost:** Another gradient boosting method, optimized for handling categorical features efficiently.

A standardized pipeline was created using scikit-learn to automate model training and evaluation.

5. Model Evaluation

The performance of each model was evaluated using accuracy.

Comparative analysis showed that ensemble methods like **XGBoost**, **ADABOOST** and **Random Forest** outperformed other models, achieving an overall prediction accuracy of 96%. These models were then used to generate match predictions for the entire season.

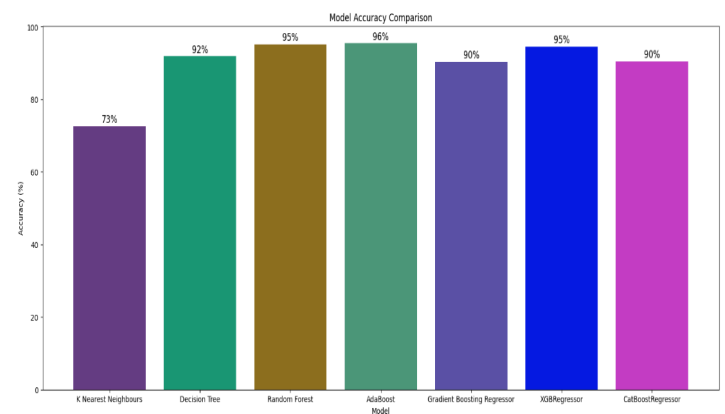
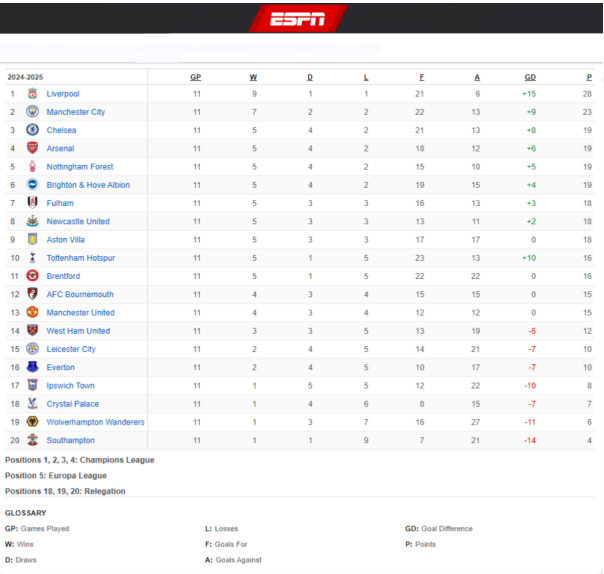


Fig 3: Model Accuracy Comparison

6. Predicting the EPL Points Table

After model evaluation, the best-performing model was used to predict the results of remaining fixtures in the current EPL season:

- **Simulating Future Matches:** Based on each team's form, statistical strengths, and fixture list, the model predicted the outcomes of future matches.
- **Points Calculation:** Match results (win, draw, loss) were translated into points (3 points for a win, 1 point for a draw, and 0 for a loss). The cumulative points were used to construct the predicted league table.
- **Table Ranking:** Teams were ranked based on total points, goal difference, and goals scored to provide a comprehensive points table projection.



The image shows a screenshot of the ESPN website displaying the actual EPL Matchweek 10 points table. The table lists 20 teams, their positions, games played (GP), wins (W), draws (D), losses (L), goals for (F), goals against (A), goal difference (GD), and points (P). Liverpool is at the top with 28 points, followed by Manchester City with 23 points. Southampton is at the bottom with 4 points.

Position	Team	GP	W	D	L	F	A	GD	P
1	Liverpool	11	9	1	1	21	6	+15	28
2	Manchester City	11	7	2	2	22	13	+9	23
3	Chelsea	11	5	4	2	21	13	+8	19
4	Arsenal	11	5	4	2	18	12	+6	19
5	Nottingham Forest	11	5	4	2	15	10	+5	19
6	Brighton & Hove Albion	11	5	4	2	19	15	+4	19
7	Fulham	11	5	3	3	16	13	+3	18
8	Newcastle United	11	5	3	3	13	11	+2	18
9	Aston Villa	11	5	3	3	17	17	0	18
10	Tottenham Hotspur	11	5	1	5	23	13	+10	16
11	Brentford	11	5	1	5	22	22	0	16
12	AFC Bournemouth	11	4	3	4	15	15	0	15
13	Manchester United	11	4	3	4	12	12	0	15
14	West Ham United	11	3	3	5	13	19	-6	12
15	Leicester City	11	2	4	5	14	21	-7	10
16	Everton	11	2	4	5	10	17	-7	10
17	Ipswich Town	11	1	5	5	12	22	-10	8
18	Crystal Palace	11	1	4	6	8	15	-7	7
19	Wolverhampton Wanderers	11	1	3	7	16	27	-11	6
20	Southampton	11	1	1	9	7	21	-14	4

Fig 4: Actual Points Table of EPL Matchweek 10

	Team	Season	Predicted Points AdaBoost
0	Liverpool	2024	28
1	Manchester City	2024	23
2	Nottingham Forest	2024	19
3	Brighton and Hove Albion	2024	19
4	Chelsea	2024	18
5	Arsenal	2024	18
6	Fulham	2024	18
7	Aston Villa	2024	18
8	Tottenham Hotspur	2024	16
9	Brentford	2024	16
11	Newcastle United	2024	15
10	Bournemouth	2024	15
12	Manchester United	2024	12
13	West Ham United	2024	12
14	Leicester City	2024	10
15	Everton	2024	10
16	Crystal Palace	2024	7
17	Wolverhampton Wanderers	2024	6
18	Ipswich Town	2024	5
19	Southampton	2024	4

Fig 5: Points Predicted by AdaBoost Model

	Team	Season	Predicted Points Random Forest
0	Liverpool	2024	28
1	Manchester City	2024	23
2	Nottingham Forest	2024	19
3	Brighton and Hove Albion	2024	19
4	Chelsea	2024	18
5	Arsenal	2024	18
6	Fulham	2024	18
7	Aston Villa	2024	18
8	Tottenham Hotspur	2024	16
9	Brentford	2024	16
11	Newcastle United	2024	15
10	Bournemouth	2024	15
12	Manchester United	2024	12
13	West Ham United	2024	12
14	Leicester City	2024	10
15	Everton	2024	10
16	Crystal Palace	2024	7
17	Wolverhampton Wanderers	2024	6
18	Ipswich Town	2024	5
19	Southampton	2024	4

Fig 6: Points Predicted by Random Forest Model

	Team	Season	Predicted Points XGB
0	Liverpool	2024	28
1	Manchester City	2024	23
2	Nottingham Forest	2024	19
3	Brighton and Hove Albion	2024	19
4	Chelsea	2024	18
5	Arsenal	2024	18
6	Fulham	2024	18
7	Aston Villa	2024	18
8	Tottenham Hotspur	2024	16
9	Brentford	2024	16
11	Newcastle United	2024	15
10	Bournemouth	2024	15
12	Manchester United	2024	12
13	West Ham United	2024	12
14	Leicester City	2024	10
15	Everton	2024	10
16	Crystal Palace	2024	7
17	Wolverhampton Wanderers	2024	6
18	Ipswich Town	2024	5
19	Southampton	2024	4

Fig 7: Points Predicted by XGB Model

IV. CONCLUSION

This research aimed to predict the English Premier League (EPL) points table by utilizing machine learning techniques to forecast match outcomes. The project involved a comprehensive approach, including data collection through web scraping, extensive data preprocessing, feature engineering, and the implementation of multiple machine learning algorithms.

1. Key Findings

- **High Prediction Accuracy:** Among the five algorithms tested (KNN, Random Forest, AdaBoost, XGBoost, and CatBoost), ensemble methods like XGBoost and CatBoost demonstrated superior performance, achieving an accuracy of 96% in predicting match outcomes. These models effectively captured the complexities of football match data, including team form, home/away advantage, and historical performance.
- **Importance of Feature Engineering:** Feature engineering played a crucial role in improving model accuracy. Custom features, such as team form indicators, head-to-head records, and rolling averages of key metrics, significantly enhanced the predictive capabilities of the models.

2. Contributions and Impact

This study demonstrates the potential of machine learning in sports analytics, particularly in forecasting league outcomes. By accurately predicting match results and league standings, the model can be a valuable tool for football analysts, bettors, and clubs looking to optimize strategies and performance.

3. Limitations

While the model achieved high accuracy, there were some limitations:

- **Data Limitations:** The predictions were constrained by the availability and quality of historical data. Factors like player injuries, transfers, and real-time match conditions (e.g., weather, referee decisions) were not accounted for, which could influence match outcomes.
- **Class Imbalance:** Predicting draws proved to be more challenging due to their relatively lower occurrence compared to wins and losses. Although some techniques were applied to address class imbalance, further optimization may be required.

4. Future Work

To build upon the current project, several avenues for future research are suggested:

- **Incorporating Real-Time Data:** Future models could integrate live match data, player statistics, and external factors (e.g., weather conditions, player injuries) to enhance prediction accuracy.
- **Advanced Feature Engineering:** Utilizing advanced football metrics like Expected Goals (xG) and Expected Assists (xA) could provide deeper insights into team performance.
- **Deep Learning Models:** Exploring deep learning techniques such as Recurrent Neural Networks (RNNs) or Long Short-Term Memory (LSTM) networks could improve predictions by capturing temporal dependencies in football data.
- **Predicting Player Performance:** Expanding the model to predict individual player performance and its impact on match outcomes could add a new dimension to football analytics.

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